

Multi-tenant Notification Service

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Time limit: 48 hours from assignment. **Stack:** Spring Boot.

The Project

A multi-tenant notification service at scale supporting multiple channels (email, SMS, push, in-app), tenant-defined templates with variable substitution, scheduled and immediate sends, per-tenant rate limiting, retries with backoff on transient failures, and delivery tracking. The service should dispatch high volumes of notifications across channels concurrently using bounded worker pools, enforce per-tenant fairness and rate limits under load, and avoid duplicate deliveries on retry. Delivery state transitions and retry attempts should be persisted with an audit trail.

Roles: platform admin (manage tenants and global limits) and tenant admin (manage templates, channel configuration, and view delivery reports).

Instructions

Framing

The Product Requirement above is intentionally open-ended. You own the scoping decisions — which entities to model, which APIs to expose, which edge cases to handle, and what to leave out. Your interpretation of the requirement and the features you choose to build are themselves part of what is being evaluated. You are encouraged to interpret the requirement generously and submit a feature-rich solution — both the breadth and the depth of the features you build contribute to the evaluation. Document every meaningful assumption in the `README.md` and explain your reasoning in the recorded video.

In Scope

The following are expected in your submission:

- REST APIs covering the core flows
- Persistence to a database of your choice
- Basic role-based access control for the roles defined in the requirement
- Input validation and error handling
- Unit & Integration Tests for the core flows

Out of Scope

Do not spend time on:

- UI or frontend
 - Deployment, containerization, or CI/CD
 - Distributed systems or microservices
 - Advanced authentication (OAuth, SSO, MFA)
 - Production-grade observability, monitoring, or alerting
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What to Submit

- **GitHub repository** (mandatory)
 - Your personal Github Project repository link
 - Multiple commits are expected during the development phase
 - Must include a `README.md`
 - Must include the `Agents.md` / `Claude.md` file used during development
 - Must include the skills used during development
 - Must include all raw files used during development
- **Video recording** (maximum 10 minutes [loom video](#)) in which you explain:
 - How you approached the problem and the solution at a high level
 - The tech stack used and the reasoning behind it
 - The AI workflow used
 - The testing approach